Session – 1

Task-1.

List 5 daily actions you perform on apps like Instagram, Zomato, or Flipkart, and for each, identify whether the data generated is numerical or categorical.

Solution:

Instagram

1) No of likes on a post – Numerical
2) Choosing a post category – Categorical

Zomato

1) Giving a restaurant rating – Numerical
2) Selecting a food type - Categorical

Flipkart

1) Adding Product to cart - Categorical

Task-2.

Given this sample data from a music app: ['Pop', 'Rock', 'Jazz', 'Hip-Hop', 'Pop', 'Rock', 'Jazz', 'Pop'], classify the type of data and explain your reasoning in 2-3 lines.

Solution:

This is **categorical data**, specifically **nominal data**, because it represents different music genres rather than measurable numerical quantities. The genres have names/categories, but there is no meaningful numerical order between Pop, Rock, Jazz, and Hip-Hop.

Task-3.

Pick any one feature from an app you use (such as the number of likes on an Instagram post, or the rating of a restaurant on Zomato) and describe how statistics could help improve that feature for users.

Solution:

Consider the **number of likes on an Instagram post**. Statistics can be used to calculate the average number of likes and identify which types of posts receive the most engagement. Instagram can then use this information to recommend more relevant content to users and help creators understand what their audience prefers.

Task-4.

Suppose you are building a new playlist recommendation feature for a Spotify-like app. List 3 basic statistical terms (such as mean, median, mode, variance, etc.) that would be useful in analyzing user listening data, and explain each with a one-line example.

Solution:

4. Useful Statistical Terms for Playlist Recommendations

1. **Mean (Average):** Finds the average listening time—for example, users may listen to music for an average of **45 minutes per day**.
2. **Mode:** Finds the most frequently listened-to genre—for example, if **Pop** is played most often, the app can recommend more Pop songs.
3. **Variance:** Measures how much listening behavior differs between users—for example, some users may listen for 20 minutes while others listen for 3 hours, showing **high variance**.